

the amount of data collected by IoT devices. Advances in computer science such as AI and data mining are helping in this regard by providing standard analytical tools. Open source Big Data tools make IoT data readable.

In tandem with the emerging IoT industry, the general software industry has moved towards open source as being a key supplier of critical software components. The major open source software used in IoT systems include:

- *Operating system:* Linux
- *Big Data:* Apache Hadoop, Apache Cassandra
- *Middleware:* Apache HTTP Server, Apache Tomcat, Eclipse Jetty
- *Cloud:* Open Stack, Cloud Foundry, Kubernetes
- *Microservices:* Docker

Open source standards provide a basis for communication among IoT devices. The Open Connectivity Foundation (OCF) and the Linux Foundation, for example, are working on a project called IoTivity to provide a framework to connect billions of IoT devices.

Open source software also play a key role in the security of IoT. Developers and cyber security experts can conduct security audits, which systematically review the source code of cyber attacks.

IoT devices do face risks, which the industries producing them are generally unprepared to deal with. Time after time, we see new data violations aimed at harming IoT products which make us increasingly cautious about buying them, with good reason. IoT devices are frequently under the threat of being hacked. In 2018, 21 per cent of the companies surveyed reported data violations or cyber attacks due to unsecured IoT devices, according to a Ponemon Institute (US based) survey. Ensuring enterprise security is a major concern for businesses. In the online world, hacks are primarily focused on identity theft, credit card fraud, etc.

IoT vendors always work hard at

implementing many of the basics when building IoT devices securely. Consider the case of security cameras. In case they are used with the same compulsory password or if it is difficult to update faulty firmware in them, then there is plenty of room for security breaches.

Protecting user information in case of credit card numbers, home addresses, or possibly more sensitive details like medical records, are some of the situations in which the security of IoT devices comes into play. With several such challenges, open source technologies may offer these manufacturers a way to develop innovative and powerful software that is more secure, while keeping up with the latest developments in the industry.

Companies that develop IoT devices should follow certain steps in order to keep their customers and their data secure:

- Encrypt user data.
- Allow users to change passwords of IoT devices from time to time. This helps to reduce the chances that a hacker could reuse a single password to harness thousands of devices (or more) for their own evil purposes.
- Use proven open source components for application development. An automated SCA (software composition analysis) tool can also alert developers to newly discovered vulnerabilities as soon as they are disclosed, giving their team the time to implement the patch before hackers have an opportunity to exploit customer applications.
- Not collect more information than is needed from IoT devices.

IoT is growing rapidly, spurred by a culture that demands digital connectivity and smart devices that can be applied to all areas of life. Many people don't realise just how vulnerable they can be when making the most of the IoT. There are several security and privacy risks that pose a serious threat. Any device that shares a wireless network is inherently

at risk of a security breach – when someone gains access to your smart devices, he or she can harvest your data and manipulate or use it.

How to provide security to IoT based systems

The following steps can be adopted to secure your family of IoT devices.

Let's start with the router: The router is said to be the 'front door' to your smart home, and this is what you need to secure first. So it is advised to invest in a well-known router that comes with a higher degree of security, as standard.

Create a secondary or guest network: You can create multiple networks on your Wi-Fi router. Create networks with parental controls for children, guest networks with potential controls, or guest networks for visitors. You can create additional networks if required. By doing this, one can prevent potential hackers from accessing sensitive data, shared files, and other bits and pieces from other devices.

Check your IoT devices' settings and keep them updated: Your IoT device probably comes with default settings and you may change them if required.

Enable two-factor authentication: Two-factor authentication is an additional security layer on top of a device password that requires secondary authentication. It is a one-time code sent via email or SMS before access is granted.

Disable UPnP features: IoT devices tend to have Universal Plug in and Play (UPnP) features, enabling different devices to find and connect to one another. This may help third-party attackers.

How to provide security to data centres connected to IoT systems

The increasing digitalisation and automation of several devices deployed across different areas of modern urban environments are set to create new security challenges for many